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Overview

This section covers administrative setup of ResolveX. Configurations are accessed via **Admin > ResolveX**.

Available Configurations

ResolveX configuration is centered around **Workflows** and their components:

- **Workflows** - define an exception handling process
- **Queues** - stages where items wait for action
- **Actions** - operations that move items between queues (or to RESOLVED)

Navigation

- **Admin > ResolveX > Workflows**

Workflows

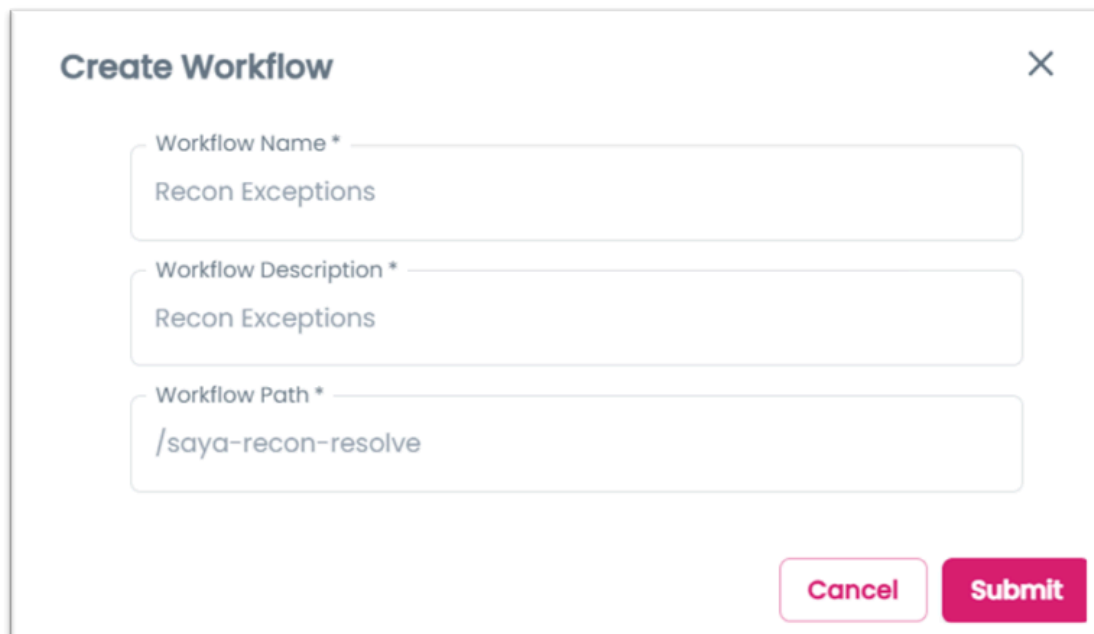
A **Workflow** defines the exception handling process - the stages (queues) and operations (actions) that items pass through.

Workflow Fields

Field	Required	Description
Process Name	Yes	Unique name for the workflow (e.g., "Payment Disputes")
Process Description	Yes	Explanation of what this workflow handles
API Context Path	No	Backend service path (defaults to <code>/saya-recon-resolve</code>)

Creating a Workflow

1. Navigate to **Admin > ResolveX > Workflows**
2. Click **Add New Workflow**
3. Enter **Process Name**
4. Enter **Process Description**
5. Optionally enter **API Context Path**

6. Click **Submit**


Create Workflow ✕

Workflow Name *
Recon Exceptions

Workflow Description *
Recon Exceptions

Workflow Path *
/saya-recon-resolve

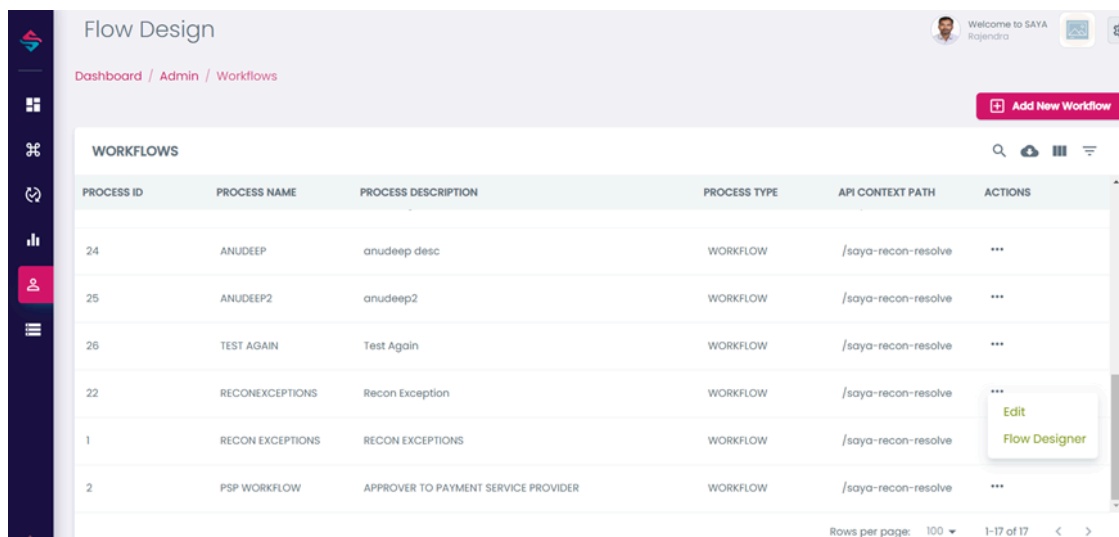
Cancel **Submit**

After Creating a Workflow

1. **Map Roles** - define which roles can access this workflow
2. **Add Queues** - create the stages items will pass through
3. **Add Actions** - define operations that move items between queues

Editing a Workflow

1. Find the workflow in the list
2. Click the **Actions** (⋮) icon
3. Select **Edit**
4. Modify fields
5. Click **Submit**



Flow Design

Welcome to SAYA Rajendra

Dashboard / Admin / Workflows

WORKFLOWS

PROCESS ID	PROCESS NAME	PROCESS DESCRIPTION	PROCESS TYPE	API CONTEXT PATH	ACTIONS
24	ANUDEEP	anudeep desc	WORKFLOW	/saya-recon-resolve	⋮
25	ANUDEEP2	anudeep2	WORKFLOW	/saya-recon-resolve	⋮
26	TEST AGAIN	Test Again	WORKFLOW	/saya-recon-resolve	⋮
22	RECONEXCEPTIONS	Recon Exception	WORKFLOW	/saya-recon-resolve	⋮ Edit Flow Designer
1	RECON EXCEPTIONS	RECON EXCEPTIONS	WORKFLOW	/saya-recon-resolve	⋮
2	PSP WORKFLOW	APPROVER TO PAYMENT SERVICE PROVIDER	WORKFLOW	/saya-recon-resolve	⋮

Rows per page: 100 1-17 of 17

Deleting a Workflow

WARNING

Cannot delete a workflow if it's used by Ops Processes. Remove from Ops Processes first.

1. Find the workflow
2. Click the **More** (...) icon
3. Select **Delete**
4. Confirm deletion

Flow Designer

The **Flow Designer** is a visual tool for building and managing workflow structure.

Accessing Flow Designer

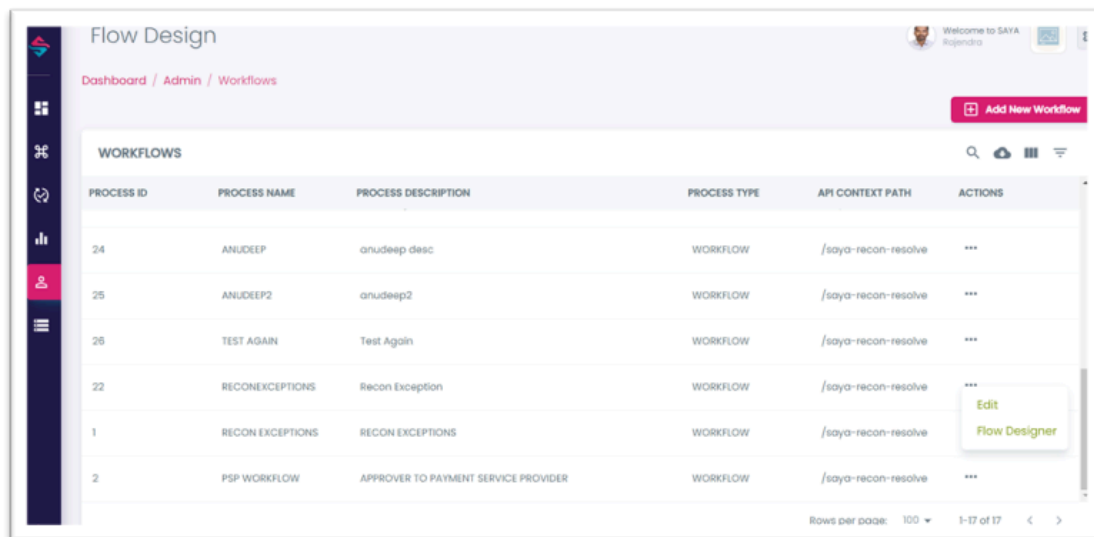
1. Navigate to **Admin > ResolveX > Workflows**
2. Click **View Flow Designer** (...)
3. Click **Actions** (...)
4. Select **Flow Designer**

Flow Designer Layout

Workflow Name (Level 1 - Root)

TEXT

```
├─ Queue 1 (Level 2)
│   ├── Action A → Target Queue
│   └── Action B → Target Queue
├─ Queue 2 (Level 2)
│   ├── Action C → Target Queue
│   └── Action D → RESOLVED
└─ Queue 3 (Level 2)
    └── Action E → RESOLVED
```

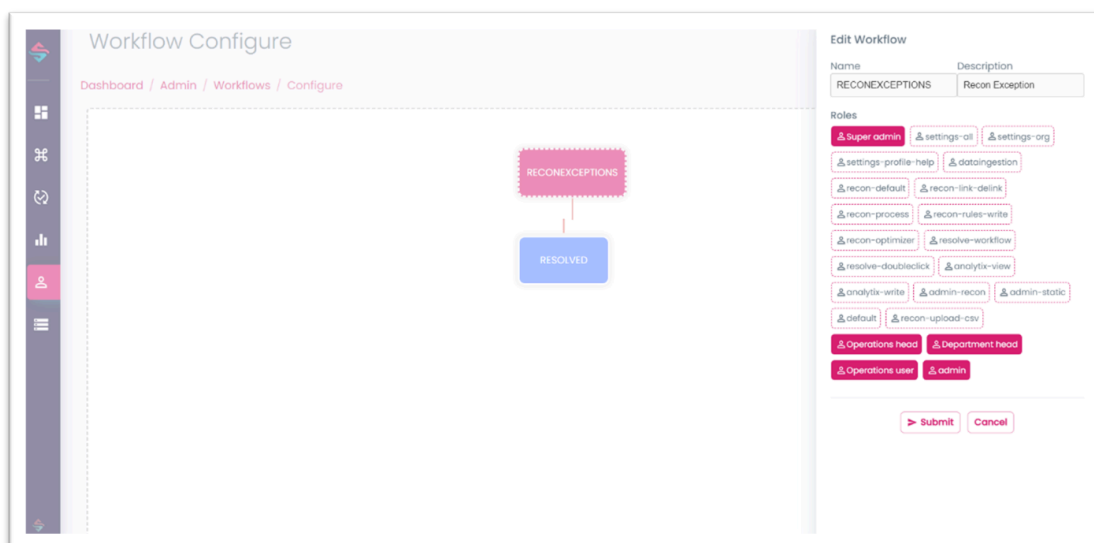


Workflow Levels

Level	Type	Description
Level 1	Workflow Root	The workflow itself - configure roles here
Level 2	Queues	Stages where items wait for action
Level 3	Actions	Operations that move items
Level 4	Target/Resolved	Final destination (another queue or RESOLVED)

Configuring Workflow Roles

1. In Flow Designer, click **Edit** (⚙️) on the root node
2. Select roles that should have access
3. Click **Save**



Only users with mapped roles can:

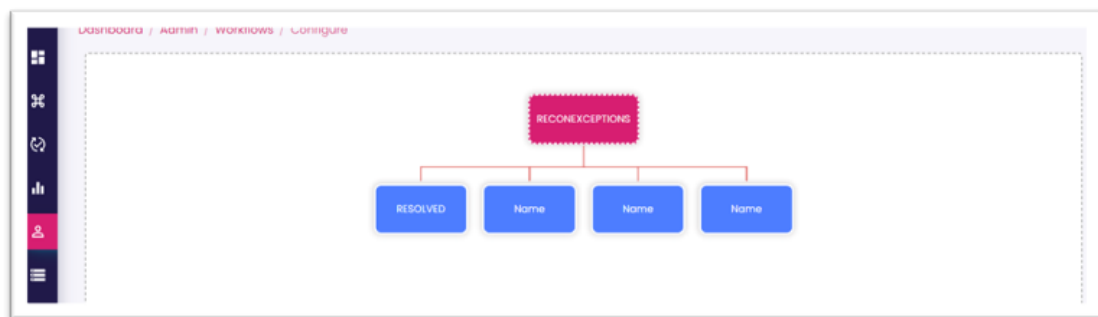
- See this workflow
- View items in the workflow's queues
- Perform actions in the workflow

Queues

Queues are stages within a workflow where items wait for action.

Adding a Queue

1. In Flow Designer, click **Add (+)** on the workflow root
2. Enter **Queue Name**
3. Enter **Queue Description**
4. Configure **Queue Roles** (subset of workflow roles)
5. Click **Save**



 Queue Roles

Queue Fields

Field	Required	Description
Queue Name	Yes	Name shown in tabs
Queue Description	No	Purpose of this stage
Queue Roles	Yes	Roles that can see items in this queue

Editing a Queue

1. Click **Edit (⚙️)** on the queue node
2. Modify name/description/roles
3. Click **Save**

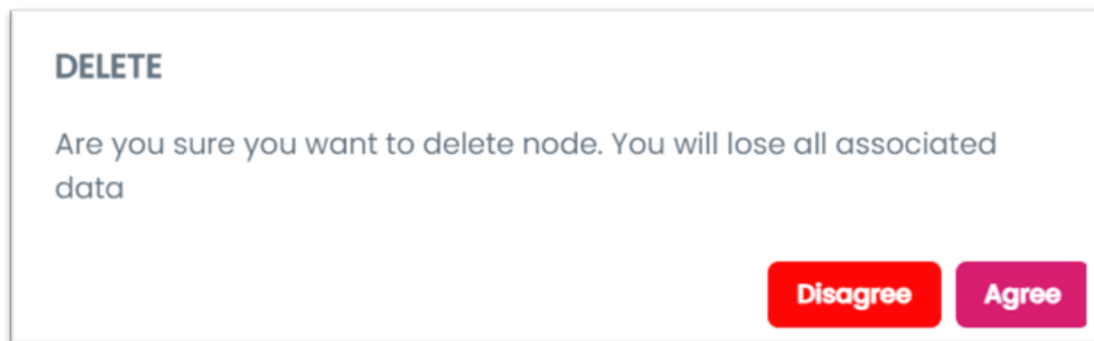
Deleting a Queue

WARNING

WARNING

Cannot delete a queue if it has items or is a target for actions.

1. Click **Remove** (–) on the queue node
2. Confirm



Reordering Queues

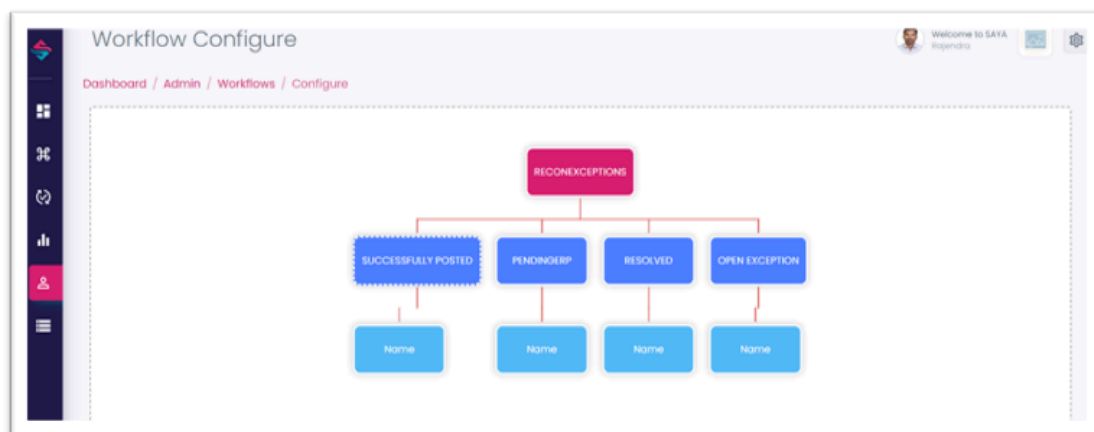
Queues can be reordered by drag-and-drop in the Flow Designer.

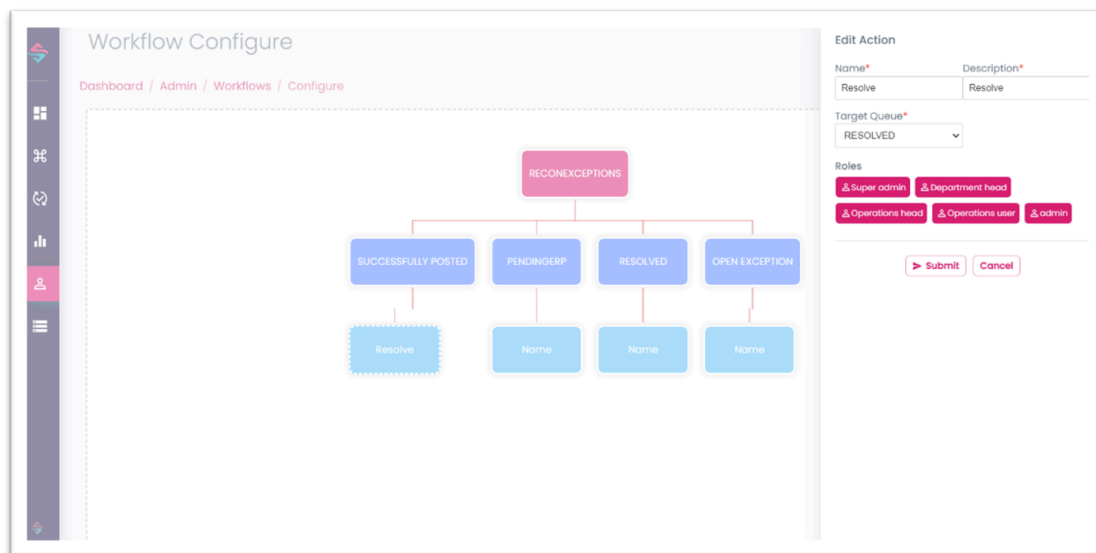
Actions

Actions are operations that move items from one queue to another.

Adding an Action

1. In Flow Designer, click **Add** (+) on a queue node
2. Enter **Action Name**
3. Enter **Action Description**
4. Select **Target Queue** (or RESOLVED)
5. Configure **Action Roles** (subset of queue roles)
6. Click **Save**





Action Fields

Field	Required	Description
Action Name	Yes	Name shown on button
Action Description	No	What the action does
Target Queue	Yes	Where item moves (or RESOLVED)
Action Roles	Yes	Roles that can perform this action

Special Target: RESOLVED

When an action's target is **RESOLVED**:

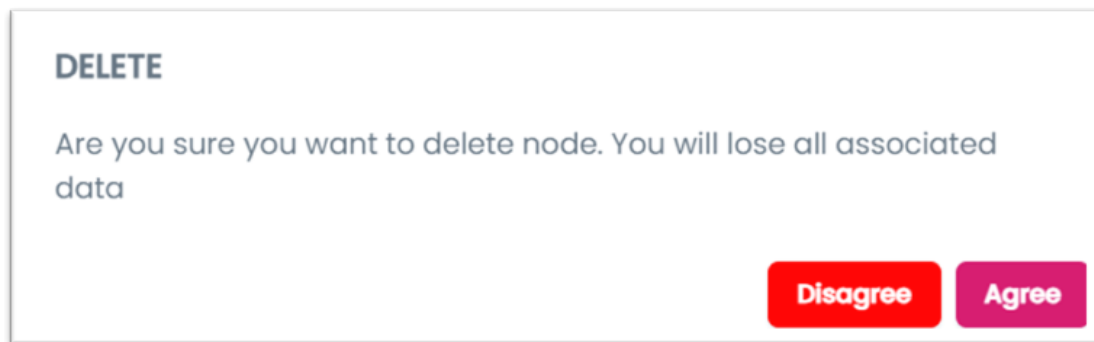
- The item is marked as complete
- It no longer appears in any queue
- It remains in the system for audit purposes

Editing an Action

1. Click **Edit** (⚙️) on the action node
2. Modify fields
3. Click **Save**

Deleting an Action

1. Click **Remove** (–) on the action node
2. Confirm



Reordering Actions

Actions within a queue can be reordered by drag-and-drop.

Role Hierarchy in Workflows

Roles are configured in a hierarchy:

```
Workflow Roles (who can see the workflow)
├── Queue Roles (subset - who can see this queue)
│   └── Action Roles (subset - who can perform this action)
```

TEXT

- A user must have the workflow role to see the workflow
- Within the workflow, they only see queues they have queue roles for
- Within a queue, they only see actions they have action roles for

Linking Workflows to Ops Processes

For a workflow to be used, it must be linked to an Ops Process:

1. Navigate to **Admin > ReconX > Ops Processes**
2. Edit an Ops Process
3. Ensure **Ops Process Type** includes "ResolveX" or "ReconX and ResolveX"
4. Select the workflow(s) from the **Workflows** dropdown
5. Save

Now items from that Ops Process can be sent to ResolveX. == Rules

Overview

Rules enable condition-based routing where workflow items automatically move between queues when specific criteria are met. Rules use FEEL (Friendly Enough Expression Language) expressions to evaluate item metadata and trigger actions without manual intervention.

When to Use Rules

- Auto-escalate high-value transactions to specialized teams
- Route items based on data thresholds (amount, score, status codes)
- Apply consistent business logic without manual triage
- Implement multi-path workflows with conditional routing

Rule Evaluation Behavior

- Rules are evaluated when an item enters a queue
- Rules are processed in priority order (lowest number = highest priority)
- **First match wins** - once a rule matches, the item moves immediately and remaining rules are skipped
- All rule evaluations are logged in the audit trail with metadata snapshots

Creating a Rule

1. Navigate to **Admin > ResolveX > Rules**
2. Click **Add New Rule**
3. Enter **Rule Name** (e.g., "High Value Auto-Escalate")
4. Enter **Rule Description** (explains business purpose)
5. Select **Metadata** (from dropdown - defines available fields for expressions)
6. Build rule expression using one of two methods:
 - a. **Rule Library** - Click existing rules to compose complex conditions using `@r.ruleId` syntax
 - b. **Direct Expression** - Type rule expression manually (autocomplete available)
7. Review **Expanded Expression** preview showing resolved references
8. Click **Submit**

ule Builder showing expression editor with library and autocomplete

Rule expression field with metadata autocomplete suggestions

Rule Expression Syntax

Rules use FEEL (Friendly Enough Expression Language) with the following syntax:

Field References

Syntax	Description	Example
<code>m.fieldName</code>	Metadata field from workflow item	<code>m.amount</code> , <code>m.transactionType</code> , <code>m.customerScore</code>
<code>s.fieldName</code>	System field (queue ID, timestamps)	<code>s.queueId</code> , <code>s.createdDate</code> , <code>s.updatedDate</code>
<code>@r.ruleId</code>	Reference to another rule for composition	<code>@r.12</code> , <code>@r.34</code>

Operators

Operator	Description	Example
>, <, >=, <=	Numeric comparison	m.amount > 100000
==, !=	Equality/inequality	m.status == "PENDING"
and, or	Logical combination	m.amount > 50000 and m.priority == "HIGH"
not	Logical negation	not(m.approved)
()	Grouping for precedence	(m.amount > 100000) or (m.vip == true)

Expression Examples

```
// Simple threshold
m.amount > 100000

// Field names with spaces or special characters
m."Utilization Limit" > 100000

// Multiple conditions
m.amount > 50000 and m.region == "APAC"

// Rule composition
@r.12 and @r.34

// Complex logic with grouping
(m.creditScore >= 700 and m.loanAmount < 50000) or m.preapproved == true

// Date comparison
s.createdDate > date("2026-01-01")
```

TEXT

Linking Rules to Workflows

1. Navigate to **Admin > ResolveX > Workflows**
2. Click **View Flow Designer** for the target workflow
3. Click **Link Rule** on the **source queue** (where items will be evaluated)
4. Click **Link New Rule**
5. Configure the rule mapping:
 - a. Select **Rule** from dropdown
 - b. Set **Priority** (lower numbers evaluated first, default: 1)
 - c. Add **Actions** - typically MOVE_QUEUE to target queue
 - d. Optionally add additional actions (SEND_EMAIL, etc.)
6. Click **Save**

 Rule link dialog showing rule selection

ow Designer showing green dashed rule edge between queues

Rule Fields

Field	Required	Description
Rule Name	Yes	Descriptive name for the rule (max 100 characters)
Rule Description	Yes	Business purpose and explanation (max 500 characters)
Metadata	Yes	Ops process metadata schema that defines available fields
Rule Expression	Yes	FEEL expression using <code>m.</code> , <code>s.</code> , <code>@r.</code> syntax
Expanded Expression	Auto-generated	Resolved expression with <code>@r.</code> references expanded recursively

Rule Mapping Fields

Field	Required	Description
Source Queue	Yes	Queue where the rule will be evaluated
Rule	Yes	Rule to evaluate against items in this queue
Priority	Yes	Evaluation order - lower numbers evaluated first (1 = highest priority)
Actions	Yes	Actions to execute when rule matches (MOVE_QUEUE, SEND_EMAIL, SEND_SMS, CALL_WEBHOOK)

Best Practices

- **Use descriptive rule names** that explain the business logic (e.g., "High Value Needs Manager Approval")
- **Set priorities carefully** to control evaluation order - most specific rules should have higher priority (lower numbers)
- **Test rules with sample data** before enabling in production workflows
- **Use rule composition** (`@r.ruleId`) to build reusable logic and reduce duplication
- **Monitor rule evaluation logs** in the audit trail to troubleshoot unexpected routing
- **Keep expressions simple** - complex rules are harder to maintain and debug
- **Document metadata field meanings** so rule authors understand available fields

Common Pitfalls

WARNING

- **Invalid field names cause runtime errors** - Field name validation happens at evaluation time, not creation time
- **Circular rule references cause infinite loops** - Rule A references Rule B which references Rule A
- **Multiple rules with same priority** have undefined evaluation order
- **Rules are not retroactive** - Only apply to items entering queue after rule is created

Events

Overview

Events enable time and schedule-based automation where actions are triggered automatically without user intervention. Events can fire based on recurring schedules (CRON) or fixed durations after item creation (TIME), making them ideal for SLA enforcement, reminders, and periodic workflows.

When to Use Events

- Escalate items sitting in a queue too long (SLA breach)
- Send daily/weekly reminder emails to users about pending items
- Trigger periodic reviews or cleanup operations
- Automate time-sensitive workflows without manual monitoring

Event Scheduling

Events use Quartz scheduler for reliable execution:

- **TIME events** - Schedule individual per-item jobs that fire once after the duration passes
- **CRON events** - Schedule recurring jobs that process all items in the queue on a schedule
- Events automatically cancel if items move to different queues
- Event execution is logged in the system audit trail

Event Types

Type	Use Case	Configuration	Behavior
CRON	Recurring schedule (daily, weekly, monthly)	Cron expression + timezone	Fires on schedule for all items in queue
TIME	Fixed duration after item creation	ISO duration (e.g., "PT2H" = 2 hours)	Fires once per item after duration passes
QUEUE_ENTER	Trigger when item enters queue	No configuration	Fires immediately (placeholder feature)

Creating an Event

1. Navigate to **Admin > ResolveX > Events**
2. Click **Add New Event**
3. Enter **Event Name** (e.g., "2 Hour SLA Escalation" or "Daily Review Reminder")
4. Enter **Event Description** explaining the business purpose
5. Select **Event Type** (CRON or TIME)
6. Configure type-specific fields:
 - a. For **CRON**:
 - i. Enter **Cron Expression** (e.g., `0 9 * * *` for 9 AM daily)
 - ii. Select **Timezone** (default: system timezone)
 - b. For **TIME**:
 - i. Use duration picker to set Days/Hours/Minutes/Seconds
 - ii. System converts to ISO duration format (e.g., "PT2H30M")
7. (Optional) Set **Starts At / Ends At** for active date window
8. Select **Metadata** (from dropdown)
9. Click **Submit**

vents CRUD screen showing list of events with types and descriptions



TIME event creation showing days/hours/minutes/seconds picker interface

Cron Expression Examples

Expression	Schedule
<code>0 9 * * *</code>	Every day at 9:00 AM
<code>0 0 * * 0</code>	Every Sunday at midnight
<code>0 */4 * * *</code>	Every 4 hours
<code>0 12 * * 1-5</code>	Weekdays (Mon-Fri) at noon
<code>0 0 1 * *</code>	First day of every month at midnight
<code>0 0 * * 1</code>	Every Monday at midnight
<code>*/15 * * * *</code>	Every 15 minutes

TIP

Use online cron expression validators like crontab.guru to test expressions before creating events.

TIME Duration Examples

ISO Duration	Meaning
PT30M	30 minutes
PT2H	2 hours
PT6H	6 hours
P1D	1 day
P1DT2H	1 day and 2 hours
P1DT2H30M	1 day, 2 hours, and 30 minutes
P7D	7 days (1 week)

NOTE

ISO durations follow ISO-8601 format: P[n]D for days, T[n]H for hours, [n]M for minutes. The UI picker generates this format automatically.

Linking Events to Workflows

1. Navigate to **Admin > ResolveX > Workflows**
2. Click **View Flow Designer** for the target workflow
3. Click **Link New Event** on the **source queue** (where items will be monitored)
4. Click **Link Event**
5. Configure the event mapping:
 - a. Select **Event** from dropdown
 - b. Choose **Action Scope**:
 - i. **ITEM** - Execute actions per-item individually (default for TIME events)
 - ii. **BATCH** - Execute actions on all items at once (default for CRON events)
 - c. (Optional) Select **Filter Rule** to limit which items receive actions
 - d. (For **BATCH** scope) Set **Max Batch Size** and **Batch Payload Type**
 - e. Add **Actions** (MOVE_QUEUE, SEND_EMAIL, SEND_SMS, CALL_WEBHOOK)
 - f. Set **Priority** if multiple events are linked to same queue
6. Click **Save**



Event link dialog for TIME event showing ITEM action scope configuration



Event link dialog for CRON event showing BATCH action scope and filter rule



Batch processing configuration showing max batch size and payload type options

Flow Designer showing amber dashed event edge with bell icon

Event Fields

Field	Required	Description
Event Name	Yes	Descriptive name for the event (max 100 characters)
Event Description	Yes	Business purpose and trigger explanation (max 500 characters)
Event Type	Yes	CRON (recurring), TIME (duration), or QUEUE_ENTER (immediate)
Event Context	Yes	Type-specific configuration (cron expression or ISO duration)
Starts At	No	Event becomes active on this date (optional date range start)
Ends At	No	Event becomes inactive after this date (optional date range end)
Metadata	Yes	Ops process metadata schema for filter rule fields

Event Mapping Fields

Field	Required	Description
Event	Yes	Event to attach to this queue
Action Scope	Yes	ITEM (per-item execution) or BATCH (all items together)
Filter Rule	No	Optional rule to filter which items receive event actions
Max Batch Size	No	Limit number of items processed per batch execution (BATCH scope only)
Batch Payload Type	No	How items are included in batch payload (BATCH scope only)
Actions	Yes	Actions to execute when event triggers (MOVE_QUEUE, SEND_EMAIL, SEND_SMS, CALL_WEBHOOK)
Priority	Yes	Evaluation order if multiple events on same queue (default: 1)

Event Behavior Details

TIME Event Lifecycle

1. Item created or enters monitored queue at 10:00 AM
2. Event mapping has PT2H (2 hour) duration
3. Quartz scheduler creates job to fire at 12:00 PM
4. At 12:00 PM:
 - a. Job checks if item is still in monitored queue
 - b. If yes: Execute actions (move queue, send email, etc.)
 - c. If no: Job completes without action (item already moved)
5. Job is removed from scheduler after execution

CRON Event Lifecycle

1. Event mapping created with `0 9 * * *` (9 AM daily)
2. Quartz scheduler creates recurring trigger
3. Every day at 9:00 AM:
 - a. Job queries all items in monitored queue
 - b. If filter rule specified: Apply filter to subset items
 - c. Execute actions based on scope:
 - i. **ITEM scope**: Loop through items, execute actions per-item
 - ii. **BATCH scope**: Group items, execute actions once with full batch
 - d. Update `lastEvaluatedAt` timestamp
4. Job continues firing daily until mapping is disabled or deleted

Action Scopes Explained

Scope	When to Use	Example
ITEM	Each item needs individual processing	Move each overdue item to escalation queue and send individual email to assigned owner
BATCH	Summary processing for multiple items	Send daily email to supervisor with list of all pending approvals (one email, multiple items)

Best Practices

- Use **descriptive event names** that explain when and why they trigger (e.g., "2 Hour SLA Escalation", "Daily 9AM Review Reminder")
- Set **active date ranges** for temporary events (e.g., holiday schedules, campaign periods)
- Use **filter rules** to target specific items (e.g., only high-priority, only unassigned)
- Test **cron expressions** with online validators before creating production events
- Monitor **scheduler logs** to confirm events are firing as expected
- Choose **appropriate action scope** - ITEM for moves, BATCH for summary notifications
- Set **reasonable batch sizes** to avoid overwhelming email systems or downstream APIs
- Consider **timezone implications** for CRON events in multi-region deployments

Common Pitfalls

WARNING

WARNING

- Invalid cron expressions accepted at creation but fail silently at runtime
- TIME events fire for moved items if no filter rule checks current queue
- CRON events with empty queues still fire (harmless but logged)
- Timezone mismatches cause events to fire at unexpected times
- No validation for future start dates - events can be created with past start dates
- Filter rules not re-evaluated for BATCH scope - evaluated once at trigger time



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